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Pricing and calibration of the futures options market: A unified approximation

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Abstract

The constant elasticity of variance (CEV) model is widely used in modeling commodity futures prices, but it may not perform well in calibrating corresponding futures options. We consider two variations of the CEV model, that is, CEV with jumps and CEV with regime switching, and compare their performance in calibrating the Chinese futures options market. In particular, we propose a unified framework for pricing American futures options by combining the continuous-time Markov chain approximation and the dynamic programming method. Results show that the inverse leverage effect in the soybean meal options market can be better described by the CEV regime-switching model.

KEYWORDS

American option, CEV model, CTMC approximation method, inverse leverage effect

1 | INTRODUCTION

The inverse leverage effect is a distinctive feature of the commodity futures market. In contrast to the leverage effect, the inverse leverage effect indicates that the changes in the returns on assets and volatility are positively correlated. Black (1976) first illustrated the leverage effect in the stock market, and showed that stock returns and volatility changes have a negative correlation. However, when it comes to the commodity market, empirical studies found that the correlation can be positive at times (see, e.g., Geman & Shih, 2008; Nomikos & Andriosopoulos, 2012). The inverse leverage effect makes sense in the commodity market since a downturn in a market will lead to a rise in the commodities price, and meanwhile it will panic the market so the volatility also increases. The inverse leverage effect varies among different commodities. For example, Geman and Shih (2008) showed the existence of the inverse leverage effect in some commodities, such as West Texas Intermediate (WTI) crude oil, copper, coal, and gold. Kristoufek (2014) estimated the volatility process for different assets and found the inverse leverage effect for natural gas. Compared with energy and metal commodities, few studies focus on the existence of the inverse leverage effect in agriculture. Since agricultural commodities are more likely to be used for hedging rather than investing, the inverse leverage effect can be more easily detected (see, e.g., Čermák, 2017; Li et al., 2017). In the options market, it has been observed that the inverse leverage effect will affect futures options prices, so that the implied volatility tends to skew rather than smile.

To investigate the correlation between price and volatility, different financial models are utilized (see, e.g., Bouchaud et al., 2001; Yu, 2005). The constant elasticity of variance (CEV) model developed by Cox (1975) is considered to be an effective model for interpreting the inverse leverage effect, and describes the relationship between the asset returns and the volatility through its elasticity parameter. Thus, by calibrating the parameters of the CEV model, the inverse leverage effect can be easily quantified. Many studies verified the inverse leverage effect by evaluating the parameters of the CEV model. For example, Geman (2009) examined the performance of the CEV model in capturing the inverse leverage effect through calibration. Čermák (2017) priced futures contracts under the CEV model, and used the Generalized Method of Moments (GMM) to calibrate the CEV model, and the calibration results have shown that both the corn and wheat commodity markets exhibited an inverse leverage effect. However, some limitations still exist in the CEV model. Though the calibration of the CEV model seems to be a standard method of detecting the inverse leverage effect, it may not explain the effect adequately well due to its simplicity.

In this paper, we consider two extensions of the CEV model: the CEV double exponential jump-diffusion (CEV-DEJD) model and the CEV regime-switching (CEV-RS) model. As an extension of the diffusion models, the jump-diffusion model combines the diffusion process with the jump process, which can better describe the price fluctuation. For example, Merton (1975) added lognormal jumps to the geometric Brownian motion, and Kou (2002) proposed the DEJD model. Another extension to the original model is the CEV-RS model. The regime-switching model can reflect the structural changes in the financial market, such as bulls and bears. Hamilton (1989) first developed the Markov regime-switching autoregressive model, and many studies focused on option pricing under the regime-switching model (see, e.g., Bollen, 1998; Boyle & Draviam, 2007; Yao et al., 2006). In this paper, we regard the CEV model as a benchmark, and quantify the results for the CEV-DEJD model and the CEV-RS model in capturing the inverse leverage effect.

To compare the performance of different models, we calibrate models to match the market prices. Since most futures options are American-style options, the task can be transformed into finding a unified approach to price American options and calibrating different models. However, it remains a challenge to price American options fast and accurately because they can be exercised at any time before the maturity date. In the last few decades, many studies focused on pricing American options under the Black–Scholes model (see, e.g., Broadie & Detemple, 1996; Myneni, 1992). Some studies also tried to solve the American option pricing problem under the CEV model (see, e.g., Detemple & Tian, 2002; Dias & Nunes, 2011).

Unlike European options, it is hard to value American options given the uncertainty of the exercise date. To solve the pricing problem, some studies tried to provide closed-form solutions to American options under specific models, while others developed numerical techniques. The former approach provides exact and explicit solutions to American options (see, e.g., Zhang & Guo, 2004; Zhao & Wong, 2012), but they are only applicable to specific models. Most of the analytical solutions focus on perpetual American options, which are not suitable for the financial industry since most futures options have specific expiration dates. Also, some numerical techniques are developed for pricing the American options under different models, which can be typically divided into three categories: the tree-based method, the finite difference method, and the simulation-based method. For the tree-based method, Maller et al. (2006) provided a multinomial approximation method for pricing American options under the general exponential Lévy process, and Beliaeva and Nawalkha (2010) generated path-independent trees under the Heston (1993) stochastic volatility model. For the finite difference method, Toivanen (2010) priced American options under jump-diffusion models through implicit finite difference discretization. Another widely used method for pricing American options is the Monte Carlo simulation (see, e.g., Glasserman, 2013; Longstaff & Schwartz, 2001). The numerical methods are generally simple to apply under different models, but it is still computationally expensive to obtain the option price. To conduct the calibration, it is crucial to find a unified pricing method that can price American options fast and accurately.

In contrast with the existing American option pricing methods we have mentioned above, the approach we developed has shown to be a unified approach. Our method can be applied to not only a specific process (such as the pure jump process or the jump-diffusion process), but also a general one-dimensional Markov process. The basic idea of our pricing method is to approximate the original Markov process by constructing a continuous-time Markov chain (CTMC), and then apply a dynamic programming algorithm to obtain the American option price. Eriksson and Pistorius (2015) provided an algorithm to evaluate American options under Markov chain models, but our paper also considers pricing American options under two-dimensional models, such as the regime-switching model. Using the algorithm proposed by Mijatović and Pistorius (2013), we can construct a CTMC to approximate the Markov process of the futures price. Numerical experiments have shown that our method can price the American option fast and accurately under one-dimensional Markov models (such as the CEV model and the CEV-DEJD model), and two-dimensional Markov models (such as the CEV-RS model).

Since our method is universally adapted, the calibration and comparison of various models can be done efficiently and accurately. This paper mainly focuses on pricing and calibrating the Chinese futures options market. To the best of our knowledge, few studies have focused on the characteristics of the Chinese futures options market. When applying the proposed method to empirical study, we calibrated the CEV, the CEV-DEJD, and the CEV-RS models to the soybean meal futures traded on the Dalian Commodity Exchange (DCE) and analyzed the calibration results. We also focused on the implied volatility curves for the soybean futures option. The calibration results show the existence of the inverse leverage effect in the Chinese futures market, and that the CEV-RS model can better describe the effect.

The remainder of this study is organized as follows: Section 2 illustrates some preliminary knowledge, including the optimal stopping problems for American options and the dynamic programming method. In Section 3, we provide the algorithm for pricing American options under the CTMC approximation method. In Section 4, we apply the CTMC approximation method to price American options under some one-dimensional models, including the CEV model and the CEV-DEJD model. Section 5 provides a way for pricing American options under two-dimensional models, such as the CEV-RS model. Section 6 analyzes the calibration results for the CTMC pricing under different models. The error analysis of our pricing method is presented in Section 7.

2 | PRELIMINARY KNOWLEDGE

This paper mainly focuses on pricing American put futures options, and the price of call options can be obtained in the same way. Let $\{\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\in[0,T]}, \mathbb{Q}\}$ be a complete probability space, where T is the time to maturity and $\mathbb{Q}$ is a risk-neutral measure. Suppose $\{X_t\}_{t\in[0,T]}$ is a time-homogeneous Markov chain, and X_t can be the economic state of any dimension at time t. For example, if we consider a one-dimensional model, such as the CEV model, then X_t is the futures price. For the two-dimensional model, such as the regime-switching model, X_t denotes the futures price and its regime. X_t is known at time t, and we can regard X_t as a random variable when the time is before t. Assume that at time t, the value of the option is a function of X_t, denoted by $f(X_t)$. We define $\mathcal{T}_{t,T}$ as the set that includes all the stopping times τ such that $t \leq \tau \leq T$. The optimal stopping time τ should maximize the expectation of the discounted payoff. $\mathbb{E}[\cdot]$ denotes the expectation taken under the risk-neutral measure $\mathbb{Q}$. The problem can be described by the Snell envelop process

$$U_t = \text{ess sup}_{\tau\in\mathcal{T}_{t,T}} \mathbb{E}[e^{-r\tau} f(X_\tau) | \mathcal{F}_t], \tag{1}$$

where r is the risk-free rate. Given $X_t = x$, we denote $V(t, x)$ as the value of an American put option at time t. Then we have

$$V(t, x) = \text{ess sup}_{\tau\in\mathcal{T}_{t,T}} \mathbb{E}_x[e^{-r(\tau-t)} f(X_\tau)]. \tag{2}$$

In this way, the American option pricing problem can be represented as an optimal stopping time problem. The value of the American option is the value obtained by exercising optimally.

The dynamic programming method can be used to estimate the optimal exercise strategy. The uniformly time-discretized Bermudan option is often used to approximate the value of the American option. Consider a Bermudan option which can be exercised only at t_i $(i = 0, 1, \ldots, M-1, M)$. When the early exercise points M for the Bermudan option become denser, the Bermudan option price converges to the corresponding American option price. The convergence can be proved by the dominated convergence theorem. For each time from t_i to the maturity date T, the option holder will follow an optimal strategy and decide whether to continue holding or exercise the option. If the payoff from early exercise at time t_i is higher than the expected payoff of the future cash flows, the option holder should exercise the option immediately at t_i, otherwise the holder should continue holding the option until the next period. If the holder decides to immediately exercise the option, the option's payoff will be $f(X_{t_i})$. Otherwise the payoff will be $\mathbb{E}_x\left[e^{-r(t_{i+1}-t_i)} V(t_{i+1}, X_{t_{i+1}})\right]$. According to Theorem 1.2.1 in Lamberton (2009), conditioning on the current economic state $X_{t_i} = x$, the value of the option at t_i is

$$V(t_i, x) = \max\left\{f(x), \mathbb{E}_x\left[e^{-r(t_{i+1}-t_i)} V(t_{i+1}, X_{t_{i+1}})\right]\right\}, \quad i = 0, 1, \ldots, M-1. \tag{3}$$

Following the iterative optimization procedure to solve Equation (3), we can derive the optimal exercise strategy and determine the option's value at each time t_i. The price of the option can be obtained by calculating $V(t, X_t)$ iteratively.

3 | PRICING AMERICAN OPTIONS UNDER THE CTMC APPROXIMATION METHOD

In this section, we construct a CTMC to approximate the futures price process, and we also implement dynamic programming algorithms to estimate the value of the American option.

A CTMC can be used to approximate the Markov process of the original model. We can construct a Markov process $\{\widetilde{X}_t\}_{0\leq t\leq T}$ to approximate the original process $\{X_t\}_{0\leq t\leq T}$. Mijatović and Pistorius (2013) developed an algorithm to construct an approximating CTMC, and the proposed algorithm constructs a CTMC in two steps. First, it creates the nonuniform state-space $\mathbf{X}$ of the CTMC, and then it defines the generator matrix $\mathbf{\Lambda}$ for the Markov chain $\{\widetilde{X}_t\}_{0 \le t \le T}$. The way to construct the transition rate matrix $\mathbf{\Lambda}$ can be different due to the original process's dimension.

Once the CTMC is constructed, we can solve the optimal stopping time problem. The price of the American option can be obtained through Equation (3). The state-space vector can be defined by $\mathbf{X} = [x_1, x_2, \ldots, x_{N-1}, x_N]^T$. Suppose that the American option can be exercised at M discrete times and set $\Delta t = T/M$. We discretized the early exercise time uniformly, that is, for $i \in \{0, 1, \ldots, M-2, M-1\}$, let $t_{i+1} = t_i + \Delta t$. The value of the option at time t_i can be expressed by $\mathbf{V}(t_i, \widetilde{X}_{t_i}) = [V(t_i, x_1), V(t_i, x_2), \ldots, V(t_i, x_{N-1}), V(t_i, x_N)]^T$. The transition rate matrix, also known as the generator matrix, is a square matrix that defines the transition rates of the CTMC. Suppose the transition rate matrix $\mathbf{\Lambda}$ has been constructed (for more details, see Sections 4 and 5). Then the transition rate matrix $\mathbf{P}$ of the CTMC can be derived from the transition rate matrix $\mathbf{\Lambda}$,

$$\mathbf{P}(\Delta t) = e^{\mathbf{\Lambda}\Delta t}. \tag{4}$$

According to Equation (3), we have

$$\begin{aligned}\mathbf{V}(t_i, \widetilde{X}_{t_i}) &= \begin{bmatrix} \max\{f(x_1), \mathbb{E}[e^{-r\Delta t}V(t_{i+1}, \widetilde{X}_{t_{i+1}})|\widetilde{X}_{t_i} = x_1]\} \\ \vdots \\ \max\{f(x_N), \mathbb{E}[e^{-r\Delta t}V(t_{i+1}, \widetilde{X}_{t_{i+1}})|\widetilde{X}_{t_i} = x_N]\} \end{bmatrix} \\ &= \max\{f(\mathbf{X}), e^{-r\Delta t}\mathbf{P}(\Delta t)\mathbf{V}(t_{i+1}, \widetilde{X}_{t_{i+1}})\},\end{aligned} \tag{5}$$

for $i = 0, 1, \ldots, M-2, M-1$, and the boundary condition is

$$\mathbf{V}(T, \widetilde{X}_T) = [f(x_0), f(x_1), \ldots, f(x_{N-1}), f(x_N)]^T = f(\mathbf{X}). \tag{6}$$

The option value for exercising immediately is denoted by $f(x)$. For example, if the price of the underlying asset follows a one-dimensional process, then $f(x) = \max(K - x, 0)$. In summary, the pricing algorithm can be implemented in two steps.

Step 1. Construct a CTMC to approximate the Markov process (see Sections 4 and 5).
1. Define a Markov chain with state-space $\mathbf{X}$.
2. Construct the transition rate matrix $\mathbf{\Lambda}$ for the state-space $\mathbf{X}$.

Step 2. Compute the American option price.
1. Obtain the boundary condition of the dynamic program. Calculate the value of the American put option at the maturity date according to Equation (6).
2. Solve Equation (5) iteratively to obtain the American option price when $t = 0$.

4 | PRICING AMERICAN OPTIONS UNDER ONE-DIMENSIONAL MODELS

In this section, we show how to price American futures options under one-dimensional models, such as the CEV model and the CEV-DEJD model. We also present some numerical experiments to test the efficiency of the CTMC approximation for the two models. All the numerical experiments in this paper were performed in Matlab 9.6.0 with a 2.4 GHz Intel Core i5 and 8 GB RAM.

4.1 | The CEV model

The CEV model was developed by Cox (1975). Under the CEV model, the futures price F_t follows the diffusion process

$$\frac{dF_t}{F_t} = \sigma\left(\frac{F_t}{F_0}\right)^{\beta} dW_t, \tag{7}$$

where $(W_u)_{u \geq t}$ is a standard Brownian motion, σ is the volatility parameter, and β is the elasticity factor. It is worth noting that when applying the CEV model to the futures price, the drift rate for the futures price is zero. This is because the process for the futures price behaves similarly as the process of the stock price with the dividend yield identical to the risk-free interest rate (see, e.g., Hull, 2014, Chap. 18).

To construct the CTMC for the CEV model, we define a Markov chain with the state-space $\mathbf{X} = \{x_1, x_2, \ldots, x_{N-1}, x_N\}$, and $\mathbf{X}$ contains the possible states of the futures price. The elements in $\mathbf{X}$ can be generated in many ways, here we follow the algorithm illustrated by Mijatović and Pistorius (2013).

The transition rate matrix of the state-space $\mathbf{X}$ can be defined by $\mathbf{\Lambda}$. Mijatović and Pistorius (2013) provided a method to obtain the transition rate matrix under the local Lévy model and we apply it to the CEV model. For the CEV model, the transition rate matrix is a tridiagonal matrix and it satisfies

$$\begin{aligned} &\sum_{j=1}^{N} \Lambda_{ij} = 0, \\ &\sum_{j=1}^{N} \Lambda_{ij}(x_j - x_i) = 0, \\ &\sum_{j=1}^{N} \Lambda_{ij}(x_j - x_i)^2 = x_i^2 \sigma(x_i)^2, \end{aligned} \tag{8}$$

where $i \neq 1, N$, the equation $\Lambda_{ij} = 0$ holds when $i = 1$ or N. For the proof of Equation (8), see Appendix A.1 for details. The solution of Equation (8) is

$$\Lambda_{ij} = \begin{cases} -\left(\dfrac{x_i}{S_0}\right)^{2\beta} \dfrac{x_i^2\sigma^2}{x_{i-1}x_i + x_{i-1}x_{i-1} - x_i x_{i+1} - x_{i-1}^2} & \text{if } j = i-1, \\ -\left(\dfrac{x_i}{S_0}\right)^{2\beta} \dfrac{x_i^2\sigma^2}{(x_{i+1} - x_{i-1})(x_i - x_{i+1})} & \text{if } j = i+1, \\ 1 - (\Lambda_{i,j-1} + \Lambda_{i,j+1}) & \text{if } j = i, \\ 0 & \text{otherwise}. \end{cases} \tag{9}$$

For the proof of Equation (9), see Appendix A.1 for details. Once the CTMC is constructed, we can use the algorithm in Section 3 to obtain the American option price.

Table 1 provides some numerical results for pricing American options under the CEV model. We compare our CTMC approximation method with known methods: the Least-Squares Monte Carlo (LSM) method proposed by Longstaff and Schwartz (2001) and the implicit finite difference method.

As shown in Table 1, the differences between the CTMC and other pricing methods are typically very small. The mean absolute difference between the CTMC and finite difference method is around 0.0020, the average absolute difference between the CTMC and the LSM method is around 0.0018.

4.2 | The CEV-DEJD model

The CEV-DEJD model is a combination of the DEJD model proposed by Kou (2002) and the CEV model. Under the CEV-DEJD model, the dynamics of the futures prices F_t follows

$$\frac{dF_t}{F_t} = -\lambda\zeta dt + \sigma\left(\frac{F_t}{F_0}\right)^{\beta} dW_t + d\left(\sum_{i=1}^{N_t}(e^{Y_i} - 1)\right), \tag{10}$$

where

$$\zeta = E[e^{Y_i}] - 1 = \frac{p\eta_1}{\eta_1 - 1} + \frac{(1-p)\eta_2}{\eta_2 + 1} - 1, \tag{11}$$

TABLE 1 Values of American put options for the CEV model

	$T = 1$ month			$T = 4$ months		
	$K = 35$	$K = 40$	$K = 45$	$K = 35$	$K = 40$	$K = 45$
$\beta = -0.5$, $\sigma = 0.2$						
Finite difference	0.008	0.842	5.000	0.223	1.570	5.060
LSM	0.008	0.850	4.997	0.225	1.567	5.058
(s.e.)	(0.000)	(0.002)	(0.001)	(0.001)	(0.003)	(0.003)
CTMC	0.008	0.849	5.000	0.223	1.570	5.058
$\beta = -0.5$, $\sigma = 0.3$						
Finite difference	0.088	1.296	5.042	0.751	2.470	5.628
LSM	0.090	1.307	5.044	0.753	2.474	5.620
(s.e.)	(0.001)	(0.002)	(0.003)	(0.002)	(0.004)	(0.005)
CTMC	0.090	1.306	5.044	0.752	2.470	5.625
$\beta = -0.125$, $\sigma = 0.2$						
Finite difference	0.007	0.842	5.000	0.203	1.571	5.078
LSM	0.007	0.850	4.996	0.205	1.574	5.076
(s.e.)	(0.000)	(0.002)	(0.001)	(0.001)	(0.003)	(0.003)
CTMC	0.007	0.849	5.000	0.204	1.571	5.076
$\beta = -0.125$, $\sigma = 0.3$						
Finite difference	0.079	1.297	5.052	0.706	2.471	5.680
LSM	0.080	1.305	5.051	0.706	2.476	5.673
(s.e.)	(0.001)	(0.002)	(0.003)	(0.002)	(0.004)	(0.005)
CTMC	0.080	1.306	5.054	0.707	2.471	5.677

Notes: We use the CTMC method to calculate the American put option prices and compare the numerical results with the solutions of the other numerical methods. The CTMC is constructed with state-space $N = 300$ and the option can be exercised at $M = 300$ discrete times. Numerical results of the implicit finite difference method are taken from Nelson and Ramaswamy (1990). For the LSM method, we suppose the option is exercisable at 50 times/year, and we use 500,000 paths to simulate the stock price process. The standard errors (s.e.) of the simulation estimates are given in parentheses. The parameter values are initial stock price $S_0 = 40$ and interest rate $r = 0.05$. It takes about 0.243 s to obtain one price via the CTMC method.

Abbreviations: CEV, constant elasticity of variance; CTMC, continuous-time Markov chain; LSM, Least-Squares Monte Carlo.

$(W_u)_{u \geq t}$ is a standard Brownian motion and $(N_u)_{u \geq t}$ is a Poisson process with rate λ. $\{Y_i, i = 1, 2, ...\}$ are independent double exponential random variables with the density function

$$f_Y(y) = p\eta_1 e^{-\eta_1 y} 1_{\{y \geq 0\}} + (1 - p)\eta_2 e^{\eta_2 y} 1_{\{y < 0\}}, \tag{12}$$

where $p \geq 0$ and $\eta_1 > 0, \eta_2 > 0$. To construct the CTMC under the CEV-DEJD model, we use the method described in Section 4.1 to obtain the state-space $\mathbf{X}$. Unlike the CEV model, the CEV-DEJD model adds a jump factor into the original model. Then the transition rate matrix $\boldsymbol{\Lambda}$ of the CTMC can be divided into $\boldsymbol{\Lambda}_{\mathrm{D}}$ and $\boldsymbol{\Lambda}_{\mathrm{J}}$, where $\boldsymbol{\Lambda}_{\mathrm{J}}$ can generate "jumps." Also, we apply the method proposed by Mijatović and Pistorius (2013) to the CEV-DEJD model. For the CEV-DEJD model, $\boldsymbol{\Lambda}_{\mathrm{J}}$ satisfies

$$\Lambda_{ij}^{\mathrm{J}} = \begin{cases} 0 & \text{if } i = 1, N, \\ -\lambda p (x_i/S_0)^{\beta}\left[(\alpha_i(j) + 1)^{-\eta_1} - (\alpha_i(j-1) + 1)^{-\eta_1}\right] & \text{if } i \neq 1, j, N; \quad \alpha_i(j-1), \alpha_i(j) > 0, \\ \lambda(1 - p)(x_i/S_0)^{\beta}\left[(\alpha_i(j) + 1)^{-\eta_2} - (\alpha_i(j-1) + 1)^{-\eta_2}\right] & \text{if } i \neq 1, j, N; \quad \alpha_i(j-1), \alpha_i(j) < 0, \\ \lambda(x_i/S_0)^{\beta}\{(1 - p)[1 - (\alpha_i(j-1) + 1)^{\eta_2}] - [p((\alpha_i(j) + 1)^{-\eta_1} - 1)]\} & \text{if } i \neq 1, j, N; \quad \alpha_i(j-1)\alpha_i(j) < 0, \\ -\sum_{\substack{j=1 \\ j \neq i}}^{N} \Lambda_{ij}^{\mathrm{J}} & \text{if } i \neq 1, N; \quad i = j, \end{cases} \tag{13}$$

where $\alpha_i(0) = -1, \alpha_i(N) = +\infty$, and we define $\alpha_i(j) = (1/2)[(x_j/x_i) - 1] + (1/2)[(x_{j+1}/x_i) - 1)]$. For the proof of Equation (13), see Appendix A.2 for details. To obtain the transition matrix $\mathbf{\Lambda}$, we also need to construct matrix $\mathbf{\Lambda_D}$. The elements of $\mathbf{\Lambda_D}$ satisfy

$$\begin{aligned}
&\sum_{j=1}^{N} \Lambda_{ij}^{\mathrm{D}} = 0,\\
&\sum_{j=1}^{N} \Lambda_{ij}^{\mathrm{D}}(x_j - x_i) = -\sum_{j=1}^{N} \Lambda_{ij}^{\mathrm{J}}(x_j - x_i),\\
&\sum_{j=1}^{N} \Lambda_{ij}^{\mathrm{D}}(x_j - x_i)^2 = x_i^2\left[\sigma(x_i)^2 + (x/S_0)^{\beta}2\lambda\left(\frac{p}{(\eta_1 - 1)(\eta_1 - 2)} + \frac{1-p}{(\eta_2 + 1)(\eta_2 + 2)}\right)\right]\\
&\qquad -\sum_{j=1}^{N} \Lambda_{ij}^{\mathrm{J}}(x_j - x_i)^2,
\end{aligned} \tag{14}$$

where $i \neq 1$ and N, and $\Lambda_{ij}^{\mathrm{D}} = 0$ when $i = 1$ or N. For the proof of Equation (14), see Appendix A.1 for details. The matrix $\mathbf{\Lambda}$ can be obtained by following the equation $\mathbf{\Lambda} = \mathbf{\Lambda_J} + \mathbf{\Lambda_D}$. Once the CTMC is constructed, we can use the algorithm in Section 3 to obtain the price of the American option and generate the optimal exercise strategy.

Table 2 provides the American put option prices under the CEV-DEJD model. The benchmark prices are computed by the LSM method proposed by Longstaff and Schwartz (2001). According to Table 2, we can see that the pricing results of the two methods are quite consistent, with an average absolute difference around 0.012.

4.3 | The early exercise boundary for the American options under a one-dimensional model

The early exercise boundary for the American option is an important part of the pricing solution. In this section, we provide a numerical algorithm to obtain the early exercise boundary through our CTMC pricing method. As mentioned in Section 2, the holder of the American option should decide whether to exercise the option before the expiration date or not. Obviously, there exists a region where the optimal strategy for the holder is to exercise the option. Also, there exists a corresponding region where the option holder should hold the option since the expected payoff is higher than early exercise. The early exercise boundary separates the two regions. The existence and uniqueness of the boundary have been demonstrated by Eriksson and Pistorius (2015). By evaluating the early exercise boundary, we can generate the optimal trading strategies for the American option. The early exercise boundary under the one-dimensional model can be obtained based on the CTMC. The algorithm for evaluating the early exercise boundary contains two steps:

Step 1. Construct a CTMC $\{\widetilde{X}_t\}_{0\le t\le T}$ with probability matrix $\mathbf{P}$ to approximate the Markov process under the one-dimensional model.

Step 2. Decide the early exercise boundary.

1. According to Equation (6), calculate the payoff of the American put option at the maturity date $f(x_T)$.
2. For $t = (M-1)\Delta t$ to 0, repeat the following steps:
   (i) For $i = 1$ to $N - 1$, compare the value between the early exercise payoff and the expected payoff. The value of matrix $\mathbf{U}$ is given by

$$\mathbf{U}(i, t) = \begin{cases} 0, & f(x_i) \le e^{-r\Delta t}\mathbf{P}(\Delta t)V(t + \Delta t, \widetilde{X}_{t+\Delta t}),\\ 1, & f(x_i) > e^{-r\Delta t}\mathbf{P}(\Delta t)V(t + \Delta t, \widetilde{X}_{t+\Delta t}). \end{cases} \tag{15}$$

   $U(i, t) = 0$ indicates that the option holder should continue holding the option and $U(i, t) = 1$ represents the option should be exercised at t.

   (ii) For $i = 1$ to $N - 1$, find i that satisfies $\mathbf{U}(i - 1, t) = 1$ and $\mathbf{U}(i, t) = 0$. Then the early exercise boundary matrix $\mathbf{B}(t) = x_i$.

TABLE 2 Pricing American put options under the CEV-DEJD model

	$\sigma = 0.2, \lambda = 7, \eta_1 = 50, \eta_2 = 25$			$\sigma = 0.3, \lambda = 3, \eta_1 = 25, \eta_2 = 50$		
	$K = 90$	$K = 100$	$K = 110$	$K = 90$	$K = 100$	$K = 110$
$\beta = -0.25$						
LSM	0.913	3.953	10.561	1.964	5.649	11.899
(s.e.)	(0.004)	(0.007)	(0.009)	(0.006)	(0.010)	(0.012)
CTMC	0.923	3.990	10.574	1.946	5.634	11.894
Abs. diff.	0.011	0.037	0.013	0.018	0.015	0.005
$\beta = -0.50$						
LSM	0.937	3.979	10.533	2.010	5.634	11.838
(s.e.)	(0.004)	(0.007)	(0.009)	(0.006)	(0.010)	(0.012)
CTMC	0.951	3.987	10.540	2.004	5.634	11.832
Abs. diff.	0.014	0.008	0.006	0.007	0.000	0.006
$\beta = -0.75$						
LSM	0.968	3.975	10.500	2.083	5.654	11.789
(s.e.)	(0.004)	(0.007)	(0.008)	(0.006)	(0.010)	(0.012)
CTMC	0.980	3.985	10.507	2.063	5.634	11.772
Abs. diff.	0.012	0.009	0.007	0.019	0.020	0.018

Notes: We compare the American put options prices obtained by the CTMC with the Monte Carlo simulation results. For the CTMC method, the size of the state-space $N = 300$ and the option can be exercised at $M = 300$ discrete times. The numerical results obtained by the LSM method are used as benchmarks. The simulation is based on 500,000 paths for the stock-price process and we suppose the option is exercisable at 50 times/year. The standard errors (s.e.) of the simulation estimates are provided. Model parameter values are the underlying stock price $S_0 = 100$ and the time to expiration of the option $T = 0.25$ year, the risk-free rate $r = 0.05$, and $p = 0.6$. It takes about 0.206 s to obtain one price via the CTMC method.

Abbreviations: CEV, constant elasticity of variance; CTMC, continuous-time Markov chain; DEJD, double exponential jump-diffusion; LSM, Least-Squares Monte Carlo.

After implementing the algorithm above, we can obtain the early exercise boundary matrix $\mathbf{B}$. For the American put option, if the price of the underlying asset at time t is higher than $\mathbf{B}(t)$, the holder should keep holding the option, otherwise the holder should exercise the option at time t.

5 | PRICING AMERICAN OPTIONS UNDER THE CEV-RS MODEL

The regime-switching model can reflect the structural changes of the economic state. We can use the regime-switching model to construct a two-dimensional Markov process $\{(R_t, F_t)\}_{0 \le t \le T}$, where F_t denotes the price of the futures and R_t denotes the regime of the futures. For example, the regime process $\{R_t\}_{0 \le t \le T}$ can have two states $R_t \in \{1, 2\}$, and the two states represent the "good" and "bad" conditions of the economy. $\{F_t\}_{0 \le t \le T}$ is a time-homogeneous Markov chain with the state-space $\{x_1, x_2, ..., x_{N-1}, x_N\}$ and transition rate matrix $\mathbf{\Lambda}$. Combining the regime-switching model with the CEV model, the price of the underlying asset F_t satisfies the stochastic differential equation

$$\frac{dF_t}{F_t} = \sigma(R_t)\left(\frac{F_t}{F_0}\right)^{\beta(R_t)} dW_t, \tag{16}$$

where $(W_u)_{u \ge t}$ is a standard Wiener process. If R_t only has one state, Equation (16) becomes the CEV model described in Section 4.1. Suppose $R_t \in \{1, 2\}$, then it has a transition probability matrix of the form

$$\begin{pmatrix} -\lambda_{11} & \lambda_{12} \\ \lambda_{21} & -\lambda_{22} \end{pmatrix}, \tag{17}$$

where $\lambda_{12}, \lambda_{21} > 0$. Given $R_t = i \ \ (i = 1, 2)$, the one-dimensional CTMC in each regime can be approximated by the algorithm described in Section 4. The transition rate matrices of CTMCs for regimes 1 and 2 can be denoted by Λ_1 and Λ_2, respectively. We can convert the original two-dimensional Markov chain into a one-dimensional Markov chain $\{\widetilde{X}_t\}_{0\le t\le T}$. According to Theorem 1 in Cai et al. (2019), the transition rate matrix of $\{\widetilde{X}_t\}_{0\le t\le T}$ can be constructed by

$$\begin{pmatrix} \lambda_{11}\mathbf{I}_N + \Lambda_1 & \lambda_{12}\mathbf{I}_N \\ \lambda_{21}\mathbf{I}_N & \lambda_{22}\mathbf{I}_N + \Lambda_2 \end{pmatrix}, \tag{18}$$

where $\mathbf{I}_N$ is the identity matrix. According to the algorithm in Section 3, we can generate the optimal exercise strategy and obtain the American option price. Since the two-dimensional regime-switching CTMC has been reduced to a one-dimensional CTMC, we can evaluate the early exercise boundary for each regime by applying the same algorithm as in Section 4.3.

Table 3 shows the comparison of option prices obtained by the CTMC method and those obtained by applying the LSM simulation under the CEV-RS model. We can see from Table 3 that the results of the CTMC method are consistent with the prices obtained from the LSM method. The average absolute difference between the two methods is 0.020.

6 | CALIBRATION

In this section, we apply our method to compare the performance of different models in calibrating the real market data. In particular, we consider the classical CEV model and its two extensions: the CEV-DEJD model and the CEV-RS model. To motivate the choice of the two extensions, we conduct empirical tests for the existence of jumps and regime switchings in Section 6.1. Then we present the calibration results in Section 6.2.

TABLE 3 Pricing American put options under the CEV-RS model

	$\sigma_1 = 0.2, \sigma_2 = 0.2, \lambda_{12} = 1, \lambda_{21} = 2$			$\sigma_1 = 0.2, \sigma_2 = 0.3, \lambda_{12} = 2, \lambda_{21} = 1$		
	$K = 90$	$K = 100$	$K = 110$	$K = 90$	$K = 100$	$K = 110$
$\beta_1 = -0.5, \beta_2 = -0.25$						
LSM	2.583	6.053	11.779	3.972	7.745	13.270
(s.e.)	(0.007)	(0.011)	(0.012)	(0.010)	(0.013)	(0.016)
CTMC	2.588	6.062	11.797	3.985	7.774	13.327
Abs. diff.	0.004	0.009	0.018	0.014	0.029	0.056
$\beta_1 = -0.5, \beta_2 = -0.5$						
LSM	2.605	6.036	11.756	4.041	7.763	13.197
(s.e.)	(0.007)	(0.010)	(0.012)	(0.010)	(0.014)	(0.016)
CTMC	2.606	6.058	11.772	4.049	7.761	13.237
Abs. diff.	0.001	0.022	0.016	0.008	0.002	0.041
$\beta_1 = -0.5, \beta_2 = -0.75$						
LSM	2.620	6.045	11.724	4.111	7.735	13.065
(s.e.)	(0.007)	(0.011)	(0.012)	(0.011)	(0.014)	(0.016)
CTMC	2.625	6.054	11.748	4.117	7.751	13.152
Abs. diff.	0.005	0.009	0.025	0.006	0.015	0.087

Notes: For the CTMC method, we set the size of the state-space $N = 300$ and the option can be exercised at $M = 100$ discrete times. For the LSM method, we suppose the option is exercisable at 50 times/year, and use 500,000 paths for the simulation of the stock price processes. The standard errors of the simulation (s.e.) are given. The model parameter values are shown as follows: the underlying stock price $S_0 = 100$, and the time to expiration of the option $T = 1$ year, and the risk-free rate $r = 0.05$. The initial state of the stock price is chosen to be in regime 1. It takes about 0.685 s to obtain one price via the CTMC method.

Abbreviations: CEV, constant elasticity of variance; CTMC, continuous-time Markov chain; LSM, Least-Squares Monte Carlo; RS, regime switching.

TABLE 4 The jump testing rejection percentage for soybean meal futures

Futures	1 min 0.05	1 min 0.10	5 min 0.05	5 min 0.10	10 min 0.05	10 min 0.10	15 min 0.05	15 min 0.10
M1801	4.49	11.89	4.08	6.91	5.71	8.54	4.49	6.10
M1805	5.31	9.35	5.31	10.16	7.35	12.20	7.35	10.16
M1807	1.63	11.84	2.86	12.20	3.67	15.45	4.08	14.23
M1808	1.22	15.38	0.41	15.38	0.41	15.79	0.41	16.19
M1809	6.91	13.36	8.13	10.53	5.31	10.12	6.94	9.72
M1811	5.71	13.41	4.08	13.01	4.08	13.41	2.86	11.38
M1812	0.41	17.89	0.41	17.48	1.63	19.51	2.12	19.11
M1901	9.84	14.75	6.56	11.43	6.17	11.43	4.49	7.35
M1903	1.64	4.49	4.10	7.35	4.51	8.98	4.90	8.16
M1905	6.10	11.74	4.47	6.88	8.13	10.12	6.50	9.31
M1907	1.65	7.79	2.47	7.79	5.35	10.66	5.71	9.43
M1908	1.23	11.48	0.82	11.84	1.63	12.24	2.04	13.88
M1909	6.58	14.75	1.65	6.56	5.35	9.43	4.49	8.61
M1911	2.05	7.38	1.64	7.35	4.90	11.43	2.86	9.39
M1912	0.82	13.11	0.41	11.89	2.45	13.11	2.86	13.11

Notes: This table shows the percentage of the days identified as having jumps by the Aït-Sahalia and Jacod (ASJ) test under different time intervals. For each futures contract, the ASJ jump test is conducted repeatedly by using daily data. The reject percentage is defined as the occurrences of the jumps per unit of time. It is calculated through the number of days we reject the null hypothesis divided by the number of the trading days. The data come from the soybean meal futures traded at the Dalian Commodity Exchange with the maturity dates from December 7, 2017 to November 7, 2019. Jumps are tested at $\alpha = 0.05$ and 0.10 significance levels, and we use the default parameters suggested by Aït-Sahalia and Jacod (2009).

6.1 | The jump test and the regime-switching test

We use the trading data of the soybean meal futures at the Dalian Commodity Exchange (Wind Data Service, 2019) to test the presence of jumps. Aït-Sahalia and Jacod (2009) provided an effective method to test jumps in a discretely observed process, which is also known as the "ASJ Jump Test." In this paper, we use the R package "highfrequency" to conduct the ASJ jump test.

We choose all of the soybean meal futures traded at the Dalian Commodity Exchange with the maturity dates from December 7, 2017 to November 7, 2019. For each trading day, we collect the prices for each futures contract and conduct the ASJ jump test: the null hypothesis is that there is no jump in the futures price for the entire day, and the alternative hypothesis is that at least one jump occurs on the testing day. We conduct the test with data collected with varying frequencies (1, 5, 10, and 15 min) and two different significance levels (0.05 and 0.10).

Table 4 shows the percentage of days that are identified as having jumps by the ASJ jump test. For example, among all of the trading days of the futures contract M1801, the proportion of days we reject the null hypothesis is 4.49% under the 0.05 significance level, and 11.89% under the 0.10 significance level. We can see from Table 4 that under the 0.05 and 0.10 significance levels, jumps in prices can be detected explicitly among all the futures in the soybean meal futures market. As the classical CEV model assumes the price process is continuous in time, the above results suggest that the model performance could be improved by adding jumps.

As an alternative, we can also extend the classical CEV model by adding the structure of regime switching. To test whether there is statistical evidence for the existence of regime switchings in the futures price, we use the method proposed by Cho and White (2007). This method proposed a quasi-likelihood-ratio (QLR) test, where the test statistic follows a function of a Gaussian process. However, the critical value cannot be obtained directly. Carter and Steigerwald (2013) simplified the procedure, and they also provided critical values for the QLR test. The null hypothesis of the QLR test is that there is only one regime (i.e., there is no regime switching), while the alternative hypothesis is that there are two regimes in the regime-switching model. The details for the QLR test can be found in Appendix B.

TABLE 5 QLR test statistics for soybean meal futures

Futures	M1801	M1803	M1805	M1807	M1808	M1809	M1811	M1812
QLR	37.46	60.29	22.95	39.87	17.20	28.58	21.34	40.86
Futures	**M1901**	**M1903**	**M1905**	**M1907**	**M1908**	**M1909**	**M1911**	**M1912**
QLR	29.54	22.11	24.00	14.69	10.57	8.20	7.70	14.73

Notes: The data come from the soybean meal futures traded at the Dalian Commodity Exchange with the maturity dates from December 7, 2017 to November 7, 2019. For each futures contract, we collect the daily closing price for all the trading days and conduct the QLR test.
Abbreviation: QLR, quasi-likelihood-ratio.

Table 5 reports the QLR test statistics for the QLR test. We can see from the table that the maximum value of the QLR is 60.29, while the minimum value of the test statistics is 7.70. Those values are far larger than the critical value 5.54 provided by Carter and Steigerwald (2013). Therefore, the null hypothesis of a single regime is rejected.

6.2 | Calibration of the soybean meal futures

To calibrate different models to the market data, we use the prices of the options on soybean meal futures traded at the Dalian Commodity Exchange from January 2, 2018 to December 28, 2018 (225 trading days). In particular, we calibrate each model on the first trading day of each week using options with different strikes and maturities. The daily closing prices of the futures options are regarded as the market prices. We use the Shanghai Interbank Offered Rate (China Foreign Exchange Trade System & National Interbank Funding Center, 2019) as the risk-free interest rate. Also, the options with a zero trading volume are eliminated.

To calibrate the model, we collect N option prices on the same trading day. The parameters values can be estimated by minimizing the sum of squared errors (SSE) between the market price and the theoretical price. For each option n ($n = 1, 2, \ldots, N-1, N$), suppose K_n is the strike price of the option n, T_n is time to expiration, and r_n is the risk-free rate corresponding to the option's maturity. Suppose $\hat{P}_n(K_n, T_n, r_n)$ is the market price, and $P_n(T_n, K_n, r_n; \theta)$ is the theoretical price calculated by the CTMC, where θ is the parameter set that needs to be estimated. We can calibrate the model by solving the following optimization problem:

$$\min_{\theta} \sum_{n=1}^{N} (\hat{P}_n(T_n, K_n, r_n) - P_n(T_n, K_n, r_n; \theta))^2. \tag{19}$$

To estimate the parameters for different models, we use the MATLAB function "fmincon" to solve Equation (19). The initial values of the parameters are selected randomly, and the calculation is repeated multiple times. We use the relative absolute error (RAE) to measure performances of different models, and RAE is given by

$$\text{RAE} = \frac{1}{N} \sum_{n=1}^{N} \left| \frac{\hat{P}_n(T_n, K_n, r_n) - P_n(T_n, K_n, r_n; \theta)}{\hat{P}_n(T_n, K_n, r_n)} \right|. \tag{20}$$

From Table 6, we can see that either adding jumps or adding regime switchings to the CEV model will lower the RAE. The calibration results also show that the CEV-RS model has the lowest mean errors, and the mean value of the RAE is 7.7%. Besides, under the CEV-RS model, the standard deviation of the RAE is also the lowest, indicating that the fitting errors in the CEV-RS model have the lowest fluctuation. In summary, the calibration results provide evidence that the CEV-RS model performs better in fitting the market price. The reason may be that the CEV-RS model can capture the complexity of the inverse leverage in the futures prices better. The elasticity parameters β for the CEV model and the CEV-DEJD model are positive, indicating that there exist inverse leverage effects in the soybean meal futures market. For the CEV-RS model, the mean value of the parameter β for regime 1 is 1.291, while for regime 2 is -1.147. Results of the CEV-RS model indicate that besides the inverse leverage effect, the soybean meal futures may also exhibit a significant leverage effect at times. Our fitted model suggests that the futures price may switch between the inverse leverage regime and the leverage regime, but the market is dominated by the inverse leverage regime: the transition rate from leverage regime to inverse leverage regime is much higher ($\lambda_{21} = 7.655$) than the other direction ($\lambda_{12} = 0.978$).

Besides, we also consider the free parameter numbers attached to each model to address the concern of over-fitting. The Akaike information criterion (AIC) and Bayesian information criterion (BIC) methods are two information criteria

TABLE 6 Parameter values and the RAE for calibrating soybean meal futures to different models

Model	CEV		CEV-DEJD		CEV-RS	
	σ	0.176	σ	0.173	σ_1	0.163
		(0.024)		(0.029)		(0.071)
	β	0.570	p	0.636	β_1	1.291
		(0.940)		(0.373)		(1.594)
			η_1	43.132	σ_2	0.163
Parameter value				(10.107)		(0.084)
			η_2	36.197	β_2	−1.147
				(14.156)		(1.997)
			λ	1.088	λ_{12}	0.978
				(2.272)		(3.333)
			β	0.777	λ_{21}	7.655
				(0.860)		(6.854)
RAE (%)		9.6		9.5		**7.7**
		(0.040)		(0.041)		(**0.038**)

Notes: This table shows the relative absolute errors (RAEs) and the mean value of the parameters for different models, the standard deviations are also provided in parentheses. For each time window, we calibrate every in-sample day's option price and calculate the RAE, respectively. The mean values and standard deviations are based on the calibration results for 50 weeks from January 2, 2018 to December 28, 2018.

Abbreviations: CEV, constant elasticity of variance; DEJD, double exponential jump-diffusion; RS, regime switching.

that can be used for model selection. In both criteria, each model is penalized with its parameter numbers. For each calibration day, we calculate the three models' AIC and BIC values. Throughout the year, there are approximately 28 weeks that the CEV-RS model outperforms other models for the AIC, and 30 weeks for the BIC, indicating that the CEV-RS model is suitable to describe the soybean meal futures market at around 56%–60% of a year.

To make a further comparison, we draw the implied volatility curves of each model on a typical day (January 5, 2018), and other days have roughly the same calibration results. In Figure 1, we can see all the implied volatility curves exhibit moneyness-related skews. For all the maturities, the CEV-RS model is closer to the actual market curve in contrast with other models. The curves of the CEV model and CEV-DEJD model almost coincide with each other for the at-the-money options. For the short-term options, the market's implied volatility curve lies between the CEV and the CEV-RS curves. For the long-term options, the three implied volatility curves all intersect at the in-the-money point. Overall, the patterns of the CEV-RS model and the market price are very close, indicating that the CEV-RS model can fit the market price accurately.

We also use the calibration results of the CEV-RS model to obtain the exercise strategy. Figure 2 provides the early exercise boundary for an option traded on January 5, 2018. We can see from Figure 2 that the two regimes have different exercise boundaries, and the exercise price for regime 2 is higher. It indicates that the option holder should exercise earlier when there exists an inverse leverage effect. This is because when the futures price falls, according to the inverse leverage effect, the volatility also decreases. So the option holder might think that the futures price is less likely to fall, and the holder will exercise the option at a higher price.

7 | ERROR ANALYSIS FOR THE CTMC PRICING METHOD

In this section, we analyze the pricing errors yielded by the CTMC American option pricing method, including the CTMC approximation errors and the dynamic programming pricing errors.

Figure 3 shows the absolute pricing errors for the CTMC pricing methods under the CEV model, the CEV-DEJD model, and the CEV-RS model. The errors in Figure 3 refer to the difference between the CTMC numerical solutions and the "true value". The errors can be represented by $\text{Error}(N) = |\text{Price}(N) - \text{Price}^*|$, where $\text{Price}(N)$ refers to the

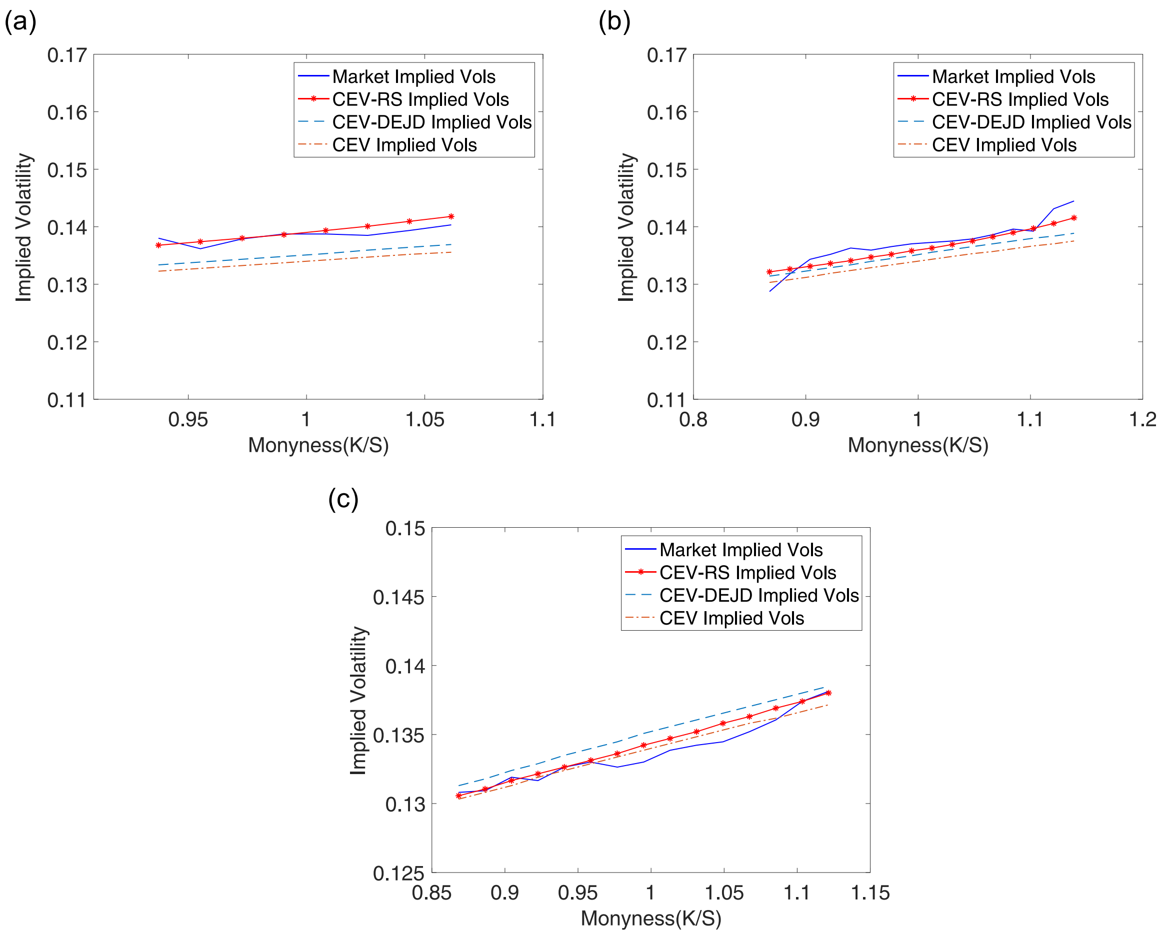

FIGURE 1 Implied volatility curves of the soybean meal futures options on January 5, 2018. (a) $T = 24$ days, (b) $T = 68$ days, and (c) $T = 153$ days. *Notes:* This figure shows the implied volatility curves of the soybean meal futures options for the CEV, the CEV-DEJD, and the CEV-RS models. The models' implied volatility curves are compared with the market-implied volatility curves. CEV, constant elasticity of variance; DEJD, double exponential jump-diffusion; RS, regime switching [Color figure can be viewed at wileyonlinelibrary.com]

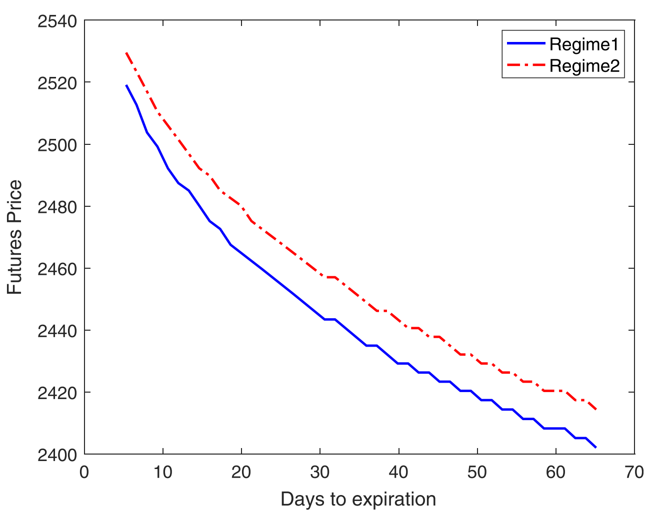

FIGURE 2 Early exercise boundaries of the soybean meal futures options under the CEV-RS model. *Notes:* We use the CEV-RS calibration results (January 5, 2018) on soybean meal futures options to evaluate the exercise boundaries in each regime. The early exercise boundary of regime 1 ($\sigma = 0.143, \beta = 0.700$) is lower than that of regime 2 ($\sigma = 0.118, \beta = -0.128$). The transition rates between the two regimes are $\lambda_{12} = 5.066$ and $\lambda_{21} = 6.758$. The risk-free rate $r = 0.0456$, the strike price $K = 2600$, futures price $F = 2766$, the option can be exercised at $M = 50$ discrete times, and the size of the state-space $N = 2000$. CEV, constant elasticity of variance; RS, regime switching [Color figure can be viewed at wileyonlinelibrary.com]

American option price derived by the CTMC method with N states and Price* represents the "true value" of the option (obtained by setting $N = 1000$). It can be seen from Figure 3 that the pricing errors are approximately linear in $\log(N)$, and the slopes of the lines vary for different models. The slopes of the lines under the CEV, CEV-DEJD, and CEV-RS models are -2.0926, -2.1207, -2.0927, respectively, indicating that the errors are approximately proportional to $1/N^2$.

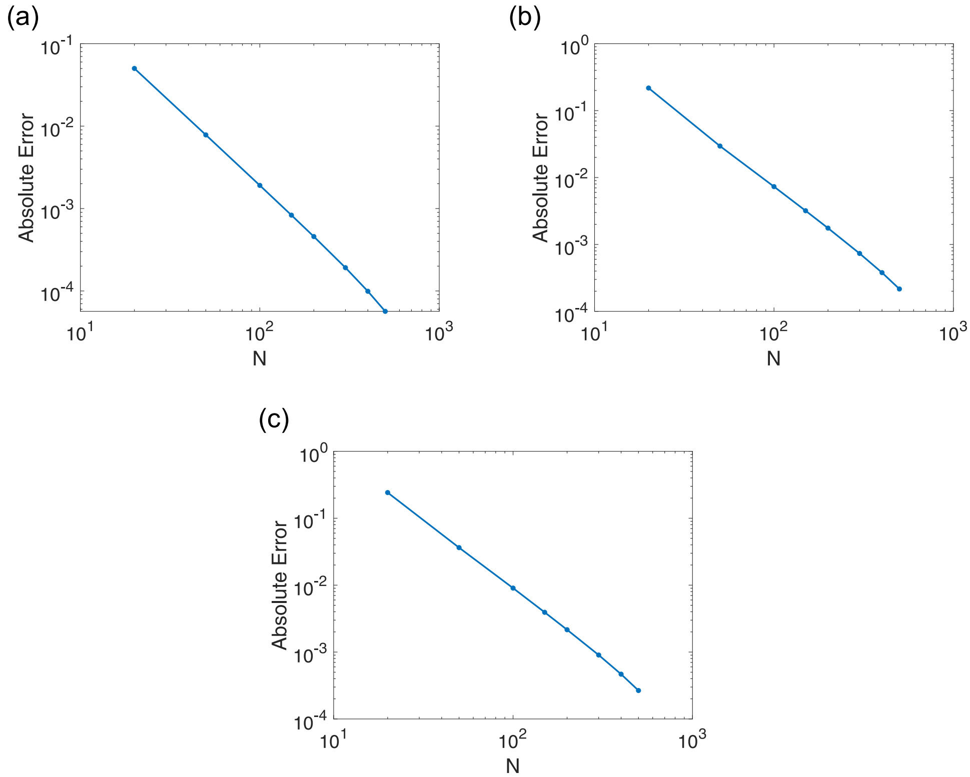

FIGURE 3 The decay of the CTMC approximation errors for pricing American options. (a) CEV model, (b) CEV-DEJD model, and (c) CEV-RS model. *Notes:* This figure shows the pricing errors for approximating American options with the CTMC method. The number of states $N = 20, 50, 100, 150, 200, 300, 400, 500$. For the CEV model, $S_0 = 40$, $K = 40$, $T = 4$ months, $r = 0.05$, $\sigma = 0.3$, $\beta = -0.5$ and the option can be exercised at $M = 1000$ discrete times. The true value of the option is obtained by using the CTMC approximation with the number of states $N = 1000$. For the DEJD model, $S_0 = 100$, $K = 100$, $T = 1$ year, $r = 0.05$, $\sigma = 0.2$, $\beta = -0.5$, $p = 0.6$, $\lambda = 7$, $\eta_1 = 50$, $\eta_2 = 25$ and the option can be exercised at $M = 1000$ discrete times. For the CEV-RS model, $S_0 = 100$, $K = 100$, $T = 1$ year, $r = 0.05$, $\sigma_1 = 0.2$, $\beta_1 = -0.5$, $\sigma_2 = 0.3$, $\beta_2 = -0.25$ and the option can be exercised at $M = 1000$ discrete times. CEV, constant elasticity of variance; CTMC, continuous-time Markov chain; DEJD, double exponential jump-diffusion; RS, regime switching [Color figure can be viewed at wileyonlinelibrary.com]

Figure 4 demonstrates the absolute errors for approximating American options with Bermuda options under the CEV model, the CEV-DEJD model, and the CEV-RS model. During each experiment, we change the exercise times M and keep the number of state N being a fixed large natural number, here we set $N = 1000$. The formulae of the errors can be defined by $\text{Error}(M) = |\text{Price}(M) - \text{Price}^*|$, where $\text{Price}(M)$ refers to the American option price derived by the CTMC method with M exercise times and Price^* represents the "true value" of the option (obtained by setting $M = 1000$). The slopes of the lines in Figure 4 vary for different models. The slopes of the lines under the CEV model, the CEV-DEJD model, and the CEV-RS model are -1.1939, -1.1942, and -1.1933, respectively, indicating that the decays of the errors are approximately at the speed of $1/M^{1.2}$.

8 | CONCLUSIONS

It is widely recognized that the futures market exhibits an inverse leverage effect, which indicates that the changes in the current futures returns and the volatility are positively correlated. The CEV model has been widely used for interpreting the inverse leverage effect. But due to the CEV model's simplicity, the inverse leverage effect may not be explained well. In this paper, we consider two variations of the classical CEV model. To compare the calibration performance under different models, we develop a unified framework for pricing the American option. The pricing method is a combination of the CTMC approximation algorithm proposed by Mijatović and Pistorius (2013) and the dynamic programming method. Numerical experiments show that our method can price American options efficiently and accurately under different models. Besides, we apply our method in calibrating the CEV model, the CEV-DEJD

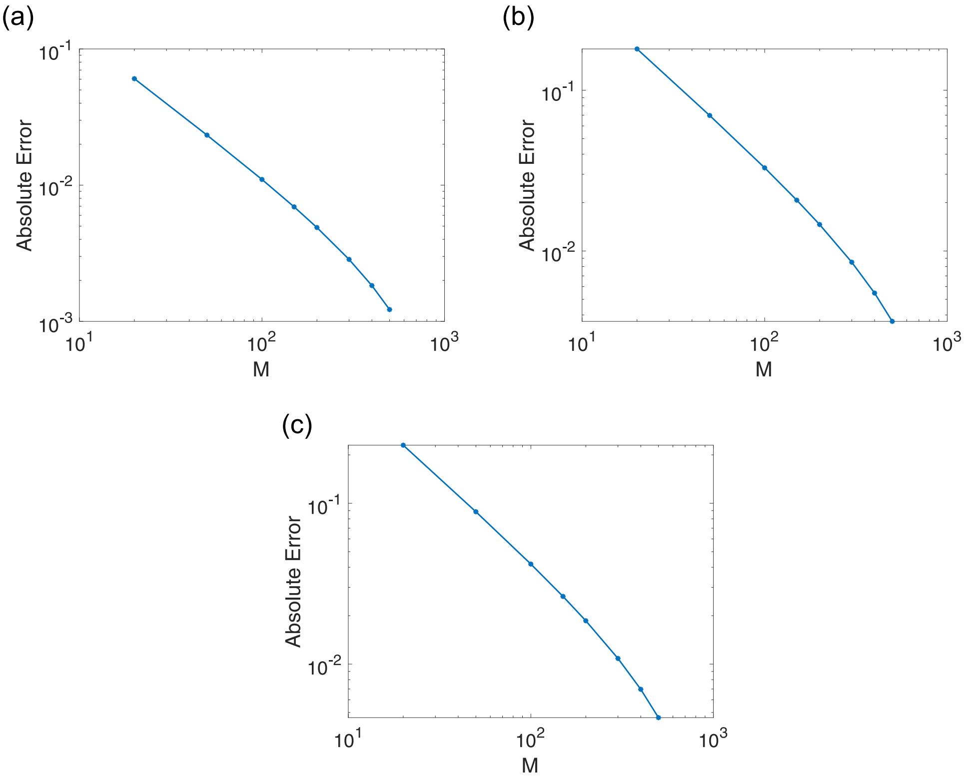

FIGURE 4 The decay of the pricing errors for approximating American options with Bermuda options. (a) CEV model, (b) CEV-DEJD model, and (c) CEV-RS model. *Notes:* This figure shows the pricing errors for approximating American options with Bermuda options under different models. The exercise time $M = 20, 50, 100, 150, 200, 300, 400, 500$. The true value of the option is obtained by using the CTMC approximation, and the option can be exercised at $M = 1000$ time points. For the CEV model, $S_0 = 40$, $K = 40$, $T = 4$ months, $r = 0.05$, $\sigma = 0.3$, $\beta = -0.5$. For the CEV-DEJD model, $S_0 = 100$, $K = 100$, $T = 1$ year, $r = 0.05$, $\sigma = 0.2$, $\beta = -0.5$, $p = 0.6$, $\lambda = 7$, $\eta_1 = 50$, $\eta_2 = 25$. For the CEV-RS model, $S_0 = 100$, $K = 100$, $T = 1$ year, $r = 0.05$, $\sigma_1 = 0.2$, $\beta_1 = -0.5$, $\sigma_2 = 0.3$, $\beta_2 = -0.25$. The number of states $N = 1000$ is set for all the options in this experiment. CEV, constant elasticity of variance; CTMC, continuous-time Markov chain; DEJD, double exponential jump-diffusion; RS, regime switching [Color figure can be viewed at wileyonlinelibrary.com]

model, and the CEV-RS model to the Chinese futures options market. Calibration results show that the inverse leverage effect can be detected by all of the three models. In addition, the calibration performance of the CEV-RS model is better than the other two models.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from Wind-Financial Terminal (http://www.wind.com.cn) and the website of Shanghai Interbank Offered Rate (SHIBOR, http://www.wind.com.cn). Restrictions apply to the availability of these data, which were used under license for this study.

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REFERENCES

Aït-Sahalia, Y., & Jacod, J. (2009). Testing for jumps in a discretely observed process. *Annals of Statistics*, *37*(1), 184–222. https://doi.org/10.2139/ssrn.1069872

Beliaeva, N. A., & Nawalkha, S. K. (2010). A simple approach to pricing American options under the Heston stochastic volatility model. *The Journal of Derivatives*, *17*(4), 25–43. https://doi.org/10.2139/ssrn.1107934

Black, F. (1976). Studies of stock price volatility changes. In *Proceedings of the Business and Economics Section of the American Statistical Association* (pp. 177–181). American Statistical Association, Alexandria, VA.

Bollen, N. P. (1998). Valuing options in regime-switching models. *The Journal of Derivatives*, *6*(1), 38–49. https://doi.org/10.3905/jod.1998.408011

Bouchaud, J. P., Matacz, A., & Potters, M. (2001). Leverage effect in financial markets: The retarded volatility model. *Physical Review Letters*, *87*(22), 228701–228704. https://doi.org/10.1103/PhysRevLett.87.228701

Boyle, P., & Draviam, T. (2007). Pricing exotic options under regime switching. *Insurance: Mathematics and Economics*, *40*(2), 267–282. https://doi.org/10.1016/j.insmatheco.2006.05.001

Broadie, M., & Detemple, J. (1996). American option valuation: New bounds, approximations, and a comparison of existing methods. *The Review of Financial Studies*, *9*(4), 1211–1250. https://doi.org/10.1093/rfs/9.4.1211

Cai, N., Kou, S., & Song, Y. (2019). A unified framework for computing regime-switching models. Available at SSRN 3310365. https://doi.org/10.2139/ssrn.3310365

Carter, A. V., & Steigerwald, D. G. (2013). Markov regime-switching tests: Asymptotic critical values. *Journal of Econometric Methods*, *2*(1), 25–34. https://doi.org/10.1515/jem-2012-0001

Čermák, M. (2017). Leverage effect and stochastic volatility in the agricultural commodity market under the CEV model. *Acta Universitatis Agriculturae et Silviculturae Mendelianae Brunensis*, *65*(5), 1671–1678. https://doi.org/10.11118/actaun201765051671

China Foreign Exchange Trade System & National Interbank Funding Center. (2019). *Shanghai interbank offered rate*. www.shibor.org

Cho, J. S., & White, H. (2007). Testing for regime switching. *Econometrica*, *75*(6), 1671–1720. https://doi.org/10.1111/j.1468-0262.2007.00809.x

Cox, J. (1975). *Notes on option pricing I: Constant elasticity of variance diffusions* (Unpublished note). Stanford University, Graduate School of Business.

Detemple, J., & Tian, W. (2002). The valuation of American options for a class of diffusion processes. *Management Science*, *48*(7), 917–937. https://doi.org/10.1287/mnsc.48.7.917.2815

Dias, J. C., & Nunes, J. P. (2011). Pricing real options under the constant elasticity of variance diffusion. *Journal of Futures Markets*, *31*(3), 230–250. https://doi.org/10.1002/fut.20468

Eriksson, B., & Pistorius, M. (2015). American option valuation under continuous-time Markov chains. *Advances in Applied Probability*, *47*(2), 378–401. https://doi.org/10.1239/aap/1435236980

Geman, H. (2009). *Commodities and commodity derivatives: Modeling and pricing for agriculturals, metals and energy*. John Wiley & Sons.

Geman, H., & Shih, Y. F. (2008). Modeling commodity prices under the CEV model. *The Journal of Alternative Investments*, *11*(3), 65–84. https://doi.org/10.3905/JAI.2009.11.3.065

Glasserman, P. (2013). *Monte Carlo methods in financial engineering*. Springer Science & Business Media.

Hamilton, J. D. (1989). A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica: Journal of the Econometric Society*, *57*(2), 357–384. https://doi.org/10.2307/1912559

Heston, S. L. (1993). A closed-form solution for options with stochastic volatility with applications to bond and currency options. *The Review of Financial Studies*, *6*(2), 327–343. https://doi.org/10.1093/rfs/6.2.327

Hull, J. C. (2014). *Options, futures, and other derivatives* (9th ed.). Pearson Prentice Hall.

Kou, S. G. (2002). A jump-diffusion model for option pricing. *Management Science*, *48*(8), 1086–1101. https://doi.org/10.2139/ssrn.242367

Kristoufek, L. (2014). Leverage effect in energy futures. *Energy Economics*, *45*, 1–9. https://doi.org/10.1016/j.eneco.2014.06.009

Lamberton, D. (2009). Optimal stopping and American options. Ljubljana Summer School on Financial Mathematics (pp. 1-43). http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.571.3408&rep=rep1&type=pdf

Li, N., Ker, A., Sam, A. G., & Aradhyula, S. (2017). Modeling regime-dependent agricultural commodity price volatilities. *Agricultural Economics*, *48*(6), 683–691. https://doi.org/10.1111/agec.12366

Longstaff, F. A., & Schwartz, E. S. (2001). Valuing American options by simulation: A simple least-squares approach. *The Review of Financial Studies*, *14*(1), 113–147. https://doi.org/10.1093/rfs/14.1.113

Maller, R. A., Solomon, D. H., & Szimayer, A. (2006). A multinomial approximation for American option prices in Lévy process models. *Mathematical Finance*, *16*(4), 613–633. https://doi.org/10.1111/j.1467-9965.2006.00286.x

Merton, R. C. (1975). Option pricing when underlying stock returns are discontinuous. *Journal of Financial Economics*, *3*(1–2), 125–144.

Mijatović, A., & Pistorius, M. (2013). Continuously monitored barrier options under Markov processes. *Mathematical Finance: An International Journal of Mathematics, Statistics and Financial Economics*, *23*(1), 1–38. https://doi.org/10.1111/j.1467-9965.2011.00486.x

Myneni, R. (1992). The pricing of the American option. *The Annals of Applied Probability*, *2*(1), 1–23. https://doi.org/10.1214/aoap/1177005768

Nelson, D. B., & Ramaswamy, K. (1990). Simple binomial processes as diffusion approximations in financial models. *The Review of Financial Studies*, *3*(3), 393–430. https://doi.org/10.1093/rfs/3.3.393

Nomikos, N., & Andriosopoulos, K. (2012). Modelling energy spot prices: Empirical evidence from NYMEX. *Energy Economics*, *34*(4), 1153–1169. https://doi.org/10.1016/j.eneco.2011.10.001

Toivanen, J. (2010). A high-order front-tracking finite difference method for pricing American options under jump-diffusion models. *The Journal of Computational Finance*, *13*(3), 61. https://doi.org/10.21314/JCF.2010.222

Wind Data Service. (2019). *Soybean meal futures options data*. www.wind.com.cn

Yao, D. D., Zhang, Q., & Zhou, X. Y. (2006). A regime-switching model for European options. In H. Yan, G. Yin, & Q. Zhang (Eds.), *Stochastic processes, optimization, and control theory: Applications in financial engineering, queueing networks, and manufacturing systems* (pp. 281–300). Springer. https://doi.org/10.1007/0-387-33815-2_14

Yu, J. (2005). On leverage in a stochastic volatility model. *Journal of Econometrics*, *127*(2), 165–178. https://doi.org/10.1016/j.jeconom.2004.08.002

Zhang, Q., & Guo, X. (2004). Closed-form solutions for perpetual American put options with regime switching. *SIAM Journal on Applied Mathematics*, *64*(6), 2034–2049. https://doi.org/10.1137/S0036139903426083

Zhao, J., & Wong, H. Y. (2012). A closed-form solution to American options under general diffusion processes. *Quantitative Finance*, *12*(5), 725–737. https://doi.org/10.1080/14697680903193405

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APPENDIX A: PROOFS

A.1 | Proofs of Equations (8), (9), and (14)

The proofs are mainly based on the studies of Mijatović and Pistorius (2013). According to Section 4.1 in Mijatović and Pistorius (2013), we can obtain the transition rate matrix Λ of the CTMC under the jump-diffusion model. Applying the methodology to the futures price, we have

$$\begin{aligned}
&\sum_{j=1}^{N}\Lambda_{ij}^{\mathrm{D}} = 0,\\
&\sum_{j=1}^{N}\Lambda_{ij}^{\mathrm{D}}(x_j - x_i) = -\sum_{j=1}^{N}\Lambda_{ij}^{\mathrm{J}}(x_j - x_i),\\
&\sum_{j=1}^{N}\Lambda_{ij}^{\mathrm{D}}(x_j - x_i)^2 = x_i^2\left[\sigma(x_i)^2 + \int_{-1}^{\infty} y^2\nu(x, dy)\right] - \sum_{j=1}^{N}\Lambda_{ij}^{\mathrm{J}}(x_j - x_i)^2,
\end{aligned} \tag{A1}$$

where Λ^{J} generates "jumps" for the CTMC, and Λ^{D} denotes the diffusion part. For the CEV-based models, we have $\sigma(x_i) = \sigma(x/S_0)^{\beta}$. Since the CEV model can be regarded as a special case of the CEV-DEJD model, we first prove the equation of the CEV-DEJD model, and then we simplify the model to prove the case under the CEV model. For the CEV-DEJD model, the Lévy measure $\nu(x, dy)$ can be defined as

$$\nu(x, dy) = (x/S_0)^{\beta}\lambda\left[p\eta_1(y+1)^{-1-\eta_1}\mathbf{I}_{(0,\infty)} + (1-p)\eta_2(y+1)^{\eta_2-1}\mathbf{I}_{(-1,0)}\right]dy, \tag{A2}$$

thus, we have

$$\int_{-1}^{\infty} y^2\nu(x, dy) = (x/S_0)^{\beta}2\lambda\left(\frac{p}{(\eta_1-1)(\eta_1-2)} + \frac{1-p}{(\eta_2+1)(\eta_2+2)}\right), \quad \eta_1 > 2. \tag{A3}$$

Combining Equations (A1) and (A3), we can obtain Equation (14).

For the CEV model, Equation (A1) can be simplified. Since the jump part of the transition rate matrix Λ^{J} is equal to 0, the transition rate matrix of the CEV model $\Lambda = \Lambda^{\mathrm{D}}$. Also, for the CEV model, $\nu(x, dy) = 0$. Following Equation (A1), we can obtain Equation (8).

To prove Equation (9), it is noteworthy that Λ of Equation (8) is a tridiagonal matrix, and $\Lambda_{ij} = 0$ when $i = 1$ or N. To solve Equation (8), we can solve the corresponding linear equations, which can be described as

$$\begin{cases}(x_{i+1}-x_i)\Lambda_{i,i+1}+(x_{i-1}-x_i)\Lambda_{i,i-1}=0,\\ (x_{i+1}-x_i)^2\Lambda_{i,i+1}+(x_{i-1}-x_i)^2\Lambda_{i,i-1}=\sigma^2 x_i^2\left(\frac{x}{S_0}\right)^{2\beta},\\ 1-(\Lambda_{i,i-1}+\Lambda_{i,i+1})=\Lambda_{i,i},\end{cases} \tag{A4}$$

where $i = 2, ..., N - 1$. By solving Equation (A4) we can obtain Equation (9).

A.2 | Proof of Equation (13)

According to Section 4.1 in Mijatović and Pistorius (2013), the jump part of the transition rate matrix can be defined as

$$\begin{cases}0 & \text{if } i=1,N,\\ \int_{\alpha_i(j-1)}^{\alpha_i(j)}\nu(x,dy) & \text{if } i\neq 1,j,N,\\ -\sum_{\substack{j=1\\ j\neq i}}^{N}\Lambda_{ij}^{\mathrm{J}} & \text{if } i\neq 1,N \text{ and } i=j.\end{cases} \tag{A5}$$

Combining Equations (A2) and (A5), we can obtain

$$\Lambda_{ij}^{\mathrm{J}}=\begin{cases}0 & \text{if } i=1,N,\\ \int_{\alpha_i(j-1)}^{\alpha_i(j)}(x/S_0)^{\beta}\lambda\left[p\eta_1(y+1)^{-1-\eta_1}\right]dy & \text{if } i\neq 1,j,N;\quad \alpha_i(j-1),\alpha_i(j)>0,\\ \int_{\alpha_i(j-1)}^{\alpha_i(j)}(x/S_0)^{\beta}\lambda\left[(1-p)\eta_2(y+1)^{\eta_2-1}\right]dy & \text{if } i\neq 1,j,N;\quad \alpha_i(j-1),\alpha_i(j)<0,\\ \int_{\alpha_i(j-1)}^{\alpha_i(j)}(x/S_0)^{\beta}\lambda\left[p\eta_1(y+1)^{-1-\eta_1}\mathbf{I}_{(0,\infty)}+(1-p)\eta_2(y+1)^{\eta_2-1}\mathbf{I}_{(-1,0)}\right]dy & \text{if } i\neq 1,j,N;\quad \alpha_i(j-1)\alpha_i(j)<0,\\ -\sum_{\substack{j=1\\ j\neq i}}^{N}\Lambda_{ij}^{\mathrm{J}} & \text{if } i\neq 1,N;\quad i=j.\end{cases} \tag{A6}$$

Computing the integrals in Equation (A6), we can obtain Equation (13).

APPENDIX B: THE DISCRETION PROCESS OF EQUATION (16) THROUGH THE EULER METHOD

Remember that $\{F_t\}_{t\in[0,T]}$ is a continuous stochastic process that follows Equation (16). We use the Euler method to discretize the original process. The Euler method generates a discrete sequence $\{F_0, F_1, ..., F_T\}$ which approximates the original process $\{F_t\}_{t\in[0,T]}$. We first separate the time interval into n equally spaced points $0 = t_0 < t_1 < \cdots < t_{n-1} < t_n = T$, then the original process can be approximated by

$$F_{n+1}-F_n=\sigma(R_n)F_n\left(\frac{F_n}{F_0}\right)^{\beta(R_n)}\Delta W_t. \tag{B1}$$

We can see from Equation (B1) that $F_{n+1} \sim N\left(F_n, \sigma^2(R_n)F_n^{2(\beta(R_n)+1)}F_0^{-2\beta(R_n)}(T/n)\right)$. Thus the log-likelihood function of the distribution can be obtained, and we can conduct the QLR test according to the method proposed by Cho and White (2007).